A resistive-strip Micromegas TPC for the X17 search at n_TOF: design, track reconstruction and performance

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The X17 search at the CERN n_TOF neutron facility requires tracking e^+e^- pairs from neutron capture on ^{3}He. To test the anomaly, their opening angle must be measured with as little material as possible. We developed a resistive-strip bulk Micromegas that works as a compact TPC: one 30 mm drift gap gives a three-dimensional track segment. Its $40 \times 40\,\mathrm{cm}^2$ area is read out through a pad matrix with a 0.78 mm pitch, capacitively coupled to two orthogonal strip views.

The resistive layer presents unique challenges for TPC operation. Charge spreads to the neighbouring strips with an RC delay, and this distorts the arrival time measured on each strip: angles taken from those times read about 4° too steep. We solve this with a novel reconstruction that fits the strip waveforms directly.

The chamber reaches an angular resolution of 1.2°, 93 % efficiency and a spatial resolution of 180 µm. At n_TOF, the reconstructed tracks image the ^{3}He target capsule, demonstrating the detector's efficacy as a TPC tracker.